

Name: **Lilian Wanjiku**

Class B Subnet Exercise

Utilize this template to develop your own IP Addressing scheme for each class B network below. The IP address of the network and the subnet sizes are provided as a starting point.

Problem # 1 – 130.6.0.0 – Subnet sizes are 8190, 16382, and 32766

Subnet Size (# of Host)	Network Address	Subnet Mask	Subnet	Increment	IP Range	Broadcast Address
32,766	130.6.0.0	/17	255.255.128.0	128	0.1-127.254	127.255
16,382	130.6.128.0	/18	255.255.192.0	64	128.1–191.254	191.255
8,190	130.6.192.0	/19	255.255.224.0	32	192.1-223.254	223.255

Problem # 2 – 152.168.0.0 – Subnet sizes are 2046, 4094, 8190, and 16382

Subnet Size (# of Host)	Network Address	Subnet Mask	Subnet	Increment	IP Range	Broadcast Address
16,382	152.168.0.0	/18	255.255.192.0	64	0.1-63.254	63.255
8,190	152.168.64.0	/19	255.255.224.0	32	64.1-95.254	95.255
4,094	152.168.96.0	/20	255.255.240.0	16	96.1-111.254	111.255
2,046	152.168.112.0	/21	255.255.248.0	8	112.1-119.254	119.255

Problem # 3 – 171.9.0.0 – Subnet sizes are 254, 510, 1022, 2046, 4094, and 8190

Subnet Size (# of Host)	Network Address	Subnet Mask	Subnet	Increment	IP Range	Broadcast Address
8,190	171.9.0.0	/19	255.255.224.0	32	0-31.254	31.255
4,094	171.9.32.0	/20	255.255.240.0	16	32.1-47.254	47.255
2,046	171.9.48.0	/21	255.255.248.0	8	48.1-55.254	55.255
1,022	171.9.56.0	/22	255.255.252.0	4	56.1-59.254	59.255
510	171.9.60.0	/23	255.255.254.0	2	60.1-61.254	61.255
254	171.9.62.0	/24	255.255.255.0	1	62.1-62.254	62..255

Problem # 4 – Make up your own given address and create 5 subnets.
 172.16.0.0 – Subnet sizes are 6, 14, 30, 62, and 126

Subnet Size (# of Host)	Network Address	Subnet Mask	Subnet	Increment	IP Range	Broadcast Address
126	172.16.0.0	/25	255.255.255.128	128	0.1–127	0.127
62	172.16.0.128	/26	255.255.255.192	64	0.128-191	0.191
30	172..16.0.192	/27	255.255.255.224	32	0.192-223	0.223
14	172.16.0.224	/28	255.255.255.240	16	0.224-239	0.239
6	172.16.0.240	/29	255.255.255.248	8	0.240-247	0.247